STA 336: Statistical Machine Learning

# Homework 4

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# Disclosure

GPT-5.3-Codex was used to create the `YAML` portion and some `LaTeX` code to format the text/equations nicely. Page formatting code was also provided by Derin Gezgin. In the setup chunk, libraries were loaded and some helper functions were defined including but not limited to `table_latex()` and `with_family()`. See the original `RMD` file here for more details.

# Problem 1

**Problem 1**

Use the full dataset and build a model using a single predictor on student. What is the mathematical form of the model?

Cell 1

```r
# Load data.
default_df <- ISLR2::Default

# Fit student-only logistic model.
model_q1 <- stats::glm(default ~ student, data = default_df, family =
stats::binomial)

# Build coefficient table.
coef_q1_tbl <- tibble::tibble(
  term     = names(stats::coef(model_q1)),
  estimate = as.numeric(stats::coef(model_q1))
) %>%
  dplyr::mutate(estimate = round(estimate, 6))

# Print coefficient table.
coef_q1_tbl %>%
  table_latex(
    col_names = c('Term', 'Estimate'),
    caption   = 'Logistic model coefficients with student only.'
  )
```

Table 1: Logistic model coefficients with student only.

| Term | Estimate |
|---|---|
| (Intercept) | -3.504128 |
| studentYes | 0.404887 |

The fitted model is

$$\log\left(\frac{\hat{p}}{1-\hat{p}}\right) = -3.504128 + 0.404887 \cdot \mathrm{I}\{\texttt{student} = \texttt{Yes}\},$$

where $\hat{p} = \Pr(\texttt{default} = \texttt{Yes} \mid \texttt{student})$. Equivalently,

$$\hat{p} = \frac{\exp(-3.504128 + 0.404887 \cdot \mathrm{I}\{\texttt{student} = \texttt{Yes}\})}{1 + \exp(-3.504128 + 0.404887 \cdot \mathrm{I}\{\texttt{student} = \texttt{Yes}\})}.$$

# Problem 2

**Problem 2**

How would you interpret the coefficient of student on the model of Problem 1?

The coefficient on `studentYes` is 0.404887, so the log-odds of default are higher for students than non-students by 0.404887 in this one-predictor model. The corresponding odds ratio is

$$\exp(0.404887) = 1.4991.$$

Therefore, the model estimates that students have about 1.50× the odds of default relative to non-students.

# Problem 3

**Problem 3**

What is the confusion matrix when you use the cutoff point to be 0.5? What is the error rate of prediction?

Cell 2

```r
# Build predictions with 0.5 cutoff.
prob_q1   <- stats::predict(model_q1, type = 'response')
pred_q1   <- factor(ifelse(prob_q1 > 0.5, 'Yes', 'No'), levels = c('No', 'Yes'))
actual_q1 <- factor(default_df$default, levels = c('No', 'Yes'))

# Build confusion matrix.
cm_q1     <- table(actual = actual_q1, predicted = pred_q1)
cm_q1_tbl <- as.data.frame.matrix(cm_q1) %>%
  tibble::rownames_to_column(var = 'Actual')

# Compute error.
error_q1 <- mean(pred_q1 != actual_q1)
error_q1_tbl <- tibble::tibble(
  metric = 'Prediction error rate',
  value  = round(error_q1, 4)
)

# Print tables.
cm_q1_tbl %>%
  table_latex(
    col_names = c('Actual', 'Predicted No', 'Predicted Yes'),
    caption   = 'Confusion matrix for student-only model (cutoff = 0.5).'
  )
```

Table 2: Confusion matrix for student-only model (cutoff = 0.5).

| Actual | Predicted No | Predicted Yes |
|---|---|---|
| No | 9667 | 0 |
| Yes | 333 | 0 |

Cell 3

```r
error_q1_tbl %>%
  table_latex(
    col_names = c('Metric', 'Value'),
    caption   = 'Error rate for student-only model (cutoff = 0.5).'
  )
```

Table 3: Error rate for student-only model (cutoff = 0.5).

| Metric | Value |
|---|---|
| Prediction error rate | 0.0333 |

At cutoff 0.5, this model predicts every observation as `No`. The prediction error rate is $\boxed{0.0333}$ (about 3.33%).

# Problem 4

**Problem 4**

Use the full dataset and build a model using a single predictor on balance. What is the mathematical form of the model?

Cell 4

```r
# Fit balance-only logistic model.
model_q4 <- stats::glm(default ~ balance, data = default_df, family =
stats::binomial)

# Build coefficient table.
coef_q4_tbl <- tibble::tibble(
  term     = names(stats::coef(model_q4)),
  estimate = as.numeric(stats::coef(model_q4))
) %>%
  dplyr::mutate(estimate = round(estimate, 6))

# Print coefficient table.
coef_q4_tbl %>%
  table_latex(
    col_names = c('Term', 'Estimate'),
    caption   = 'Logistic model coefficients with balance only.'
  )
```

Table 4: Logistic model coefficients with balance only.

| Term | Estimate |
|---|---|
| (Intercept) | -10.651331 |
| balance | 0.005499 |

The fitted model is

$$\log\left(\frac{\hat{p}}{1-\hat{p}}\right) = -10.651331 + 0.005499\,\texttt{balance},$$

where $\hat{p} = \Pr(\texttt{default} = \texttt{Yes} \mid \texttt{balance})$. Equivalently,

$$\hat{p} = \frac{\exp(-10.651331 + 0.005499\,\texttt{balance})}{1 + \exp(-10.651331 + 0.005499\,\texttt{balance})}.$$

# Problem 5

**Problem 5**

How would you interpret the coefficient of balance on the model of Problem 4?

The coefficient on `balance` is 0.005499, so each one-unit increase in balance increases the log-odds of default by 0.005499. In odds terms,

$$\exp(0.005499) = 1.0055,$$

so each \$1 increase in balance multiplies the odds of default by about 1.0055.

# Problem 6

**Problem 6**

What is the confusion matrix of the model in Problem 4 when you use the cutoff point to be 0.5? What is the error rate of prediction?

Cell 5

```r
# Build predictions with 0.5 cutoff.
prob_q4   <- stats::predict(model_q4, type = 'response')
pred_q4   <- factor(ifelse(prob_q4 > 0.5, 'Yes', 'No'), levels = c('No', 'Yes'))
actual_q4 <- factor(default_df$default, levels = c('No', 'Yes'))

# Build confusion matrix.
cm_q4 <- table(actual = actual_q4, predicted = pred_q4)
cm_q4_tbl <- as.data.frame.matrix(cm_q4) %>%
  tibble::rownames_to_column(var = 'Actual')

# Compute error.
error_q4 <- mean(pred_q4 != actual_q4)
error_q4_tbl <- tibble::tibble(
  metric = 'Prediction error rate',
  value  = round(error_q4, 4)
)

# Print tables.
cm_q4_tbl %>%
  table_latex(
    col_names = c('Actual', 'Predicted No', 'Predicted Yes'),
    caption   = 'Confusion matrix for balance-only model (cutoff = 0.5).'
  )
```

Table 5: Confusion matrix for balance-only model (cutoff = 0.5).

| Actual | Predicted No | Predicted Yes |
|---|---|---|
| No | 9625 | 42 |
| Yes | 233 | 100 |

Cell 6

```r
error_q4_tbl %>%
  table_latex(
    col_names = c('Metric', 'Value'),
    caption   = 'Error rate for balance-only model (cutoff = 0.5).'
  )
```

Table 6: Error rate for balance-only model (cutoff = 0.5).

| Metric | Value |
|---|---|
| Prediction error rate | 0.0275 |

The balance-only model has prediction error $\boxed{0.0275}$ (about 2.75%).

# Problem 7

**Problem 7**

Split the data into halves and use one half as the training and the other half as the test. Fit a model on the training data using `student`, `balance`, and `income` as predictors.

Cell 7

```r
# Create train/test split.
set.seed(20060527)
n_default     <- nrow(default_df)
train_idx     <- sample(seq_len(n_default), n_default / 2)
default_train <- default_df[train_idx, ]
default_test  <- default_df[-train_idx, ]

# Fit training model.
model_q7 <- stats::glm(
  default ~ student + balance + income,
  data   = default_train,
  family = stats::binomial
)

# Build split table.
split_q7_tbl <- tibble::tibble(
  data_set     = c('Training', 'Test'),
  observations = c(nrow(default_train), nrow(default_test))
)

# Build coefficient table.
coef_q7_tbl <- as.data.frame(stats::coef(summary(model_q7))) %>%
  tibble::rownames_to_column(var = 'term') %>%
  dplyr::rename(
    estimate  = Estimate,
    std_error = `Std. Error`,
    z_value   = `z value`,
    p_value   = `Pr(>|z|)`
  ) %>%
  dplyr::mutate(dplyr::across(-term, ~ round(.x, 6)))

# Create probabilities for later questions.
prob_train_q7 <- stats::predict(model_q7, newdata = default_train, type = 'response')
prob_test_q7  <- stats::predict(model_q7, newdata = default_test, type = 'response')

# Print tables.
split_q7_tbl %>%
  table_latex(
    col_names = c('Data set', 'Observations'),
    caption   = 'Train/test split sizes.'
```

```
  )
```

Table 7: Train/test split sizes.

| Data set | Observations |
|---|---|
| Training | 5000 |
| Test | 5000 |

Cell 8

```
coef_q7_tbl %>%
  table_latex(
    col_names = c('Term', 'Estimate', 'Std. Error', 'z value', 'Pr(>|z|)'),
    caption   = 'Logistic regression coefficient summary.'
  )
```

Table 8: Logistic regression coefficient summary.

| Term | Estimate | Std. Error | z value | Pr(>\|z\|) |
|---|---|---|---|---|
| (Intercept) | -11.362494 | 0.729852 | -15.568209 | 0.000000 |
| studentYes | -0.439823 | 0.345068 | -1.274598 | 0.202452 |
| balance | 0.005885 | 0.000340 | 17.319003 | 0.000000 |
| income | 0.000008 | 0.000012 | 0.693137 | 0.488224 |

# Problem 8

**Problem 8**

What is the training error of the model in Problem 7?

Cell 9

```r
# Build training predictions.
pred_train_q7   <- factor(ifelse(prob_train_q7 > 0.5, 'Yes', 'No'), levels =
c('No', 'Yes'))
actual_train_q7 <- factor(default_train$default, levels = c('No', 'Yes'))

# Compute training error.
train_error_q7  <- mean(pred_train_q7 != actual_train_q7)
train_error_tbl <- tibble::tibble(
  metric = 'Training error rate',
  value  = round(train_error_q7, 4)
)

# Print training error table.
train_error_tbl %>%
  table_latex(
    col_names = c('Metric', 'Value'),
    caption   = 'Training error with cutoff = 0.5.'
  )
```

Table 9: Training error with cutoff = 0.5.

| Metric | Value |
|---|---|
| Training error rate | 0.0264 |

The training error is $\boxed{0.0264}$ (about 2.64%).

# Problem 9

**Problem 9**

What is the test error of the model in Problem 7?

Cell 10

```r
# Build test predictions.
pred_test_q7   <- factor(ifelse(prob_test_q7 > 0.5, 'Yes', 'No'), levels =
c('No', 'Yes'))
actual_test_q7 <- factor(default_test$default, levels = c('No', 'Yes'))

# Compute test error.
test_error_q7  <- mean(pred_test_q7 != actual_test_q7)
test_error_tbl <- tibble::tibble(
  metric = 'Test error rate',
  value  = round(test_error_q7, 4)
)

# Print test error table.
test_error_tbl %>%
  table_latex(
    col_names = c('Metric', 'Value'),
    caption   = 'Test error with cutoff = 0.5.'
  )
```

Table 10: Test error with cutoff = 0.5.

| Metric | Value |
|---|---|
| Test error rate | 0.0266 |

The test error is $\boxed{0.0266}$ (about 2.66%).

# Problem 10

**Problem 10**

Draw the ROC curve for model in Problem 7 and find the best cutoff point. What does the best cutoff point mean?

Cell 11

```r
# Build ROC.
roc_q7 <- Epi::ROC(
  form = default ~ student + balance + income,
  data = default_train,
  plot = 'ROC',
  MX   = TRUE
)
```

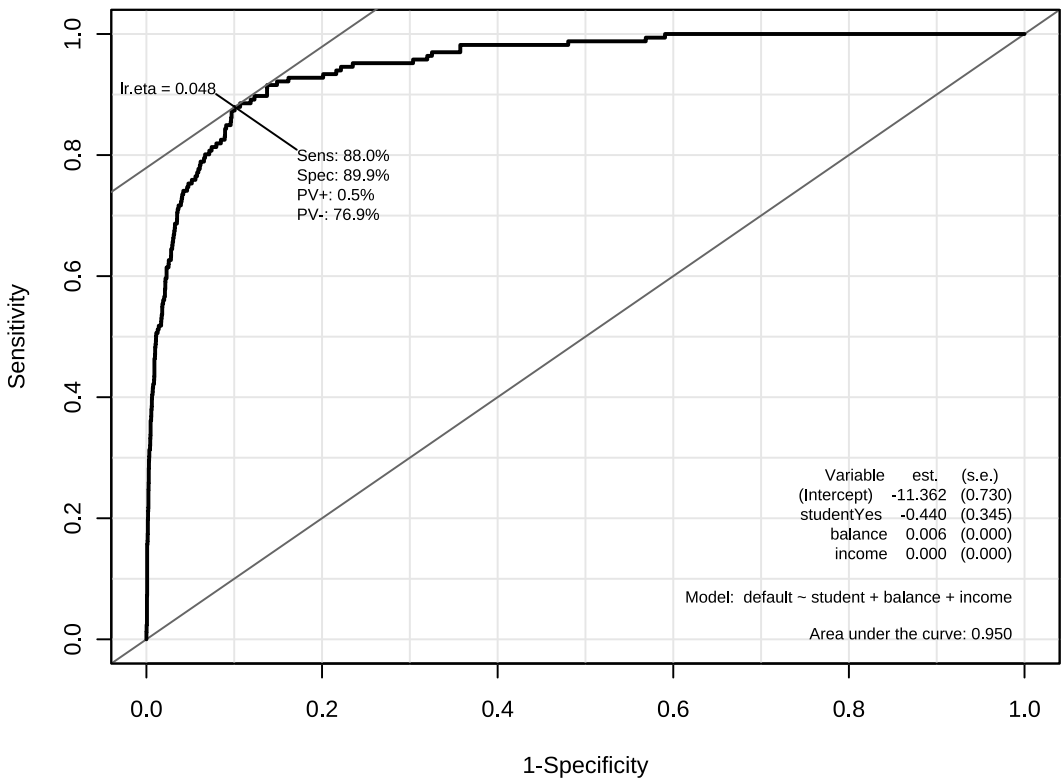


Figure 1: ROC curve on training data for the multivariable model.

Cell 12

```r
# Extract ROC grid.
roc_q7_res <- roc_q7$res

# Find best cutoff with MX criterion.
```

```r
best_row_q7 <- roc_q7_res %>%
  dplyr::mutate(mx_value = sens + spec) %>%
  dplyr::filter(mx_value == max(mx_value)) %>%
  dplyr::arrange(dplyr::desc(lr.eta)) %>%
  dplyr::slice(1)
best_cutoff_q7 <- best_row_q7$lr.eta[[1]]

# Build best-cutoff table.
best_cutoff_tbl <- tibble::tibble(
  metric = c(
    'Best cutoff',
    'Sensitivity',
    'Specificity',
    'False positive rate',
    'MX value (sensitivity + specificity)',
    'AUC'
  ),
  value  = round(c(
    best_cutoff_q7,
    best_row_q7$sens[[1]],
    best_row_q7$spec[[1]],
    1 - best_row_q7$spec[[1]],
    best_row_q7$mx_value[[1]],
    roc_q7$AUC
  ), 4)
)

# Print best-cutoff table.
best_cutoff_tbl %>%
  table_latex(
    col_names = c('Metric', 'Value'),
    caption   = 'Best cutoff chosen from training ROC.'
  )
```

Table 11: Best cutoff chosen from training ROC.

| Metric | Value |
|---|---|
| Best cutoff | 0.0477 |
| Sensitivity | 0.8795 |
| Specificity | 0.8995 |
| False positive rate | 0.1005 |
| MX value (sensitivity + specificity) | 1.7790 |
| AUC | 0.9498 |

Using (`Epi::ROC` with `MX = TRUE`), the best cutoff is $\boxed{0.0477}$. This is the threshold that maximizes sensitivity + specificity on the training data.

# Problem 11

**Problem 11**

Use the best cutoff point on the test data. What is the test error now?

Cell 13

```r
# Build test predictions with best cutoff.
pred_test_best_q7 <- factor(ifelse(prob_test_q7 > best_cutoff_q7, 'Yes', 'No'),
levels = c('No', 'Yes'))

# Compute test error with best cutoff.
test_error_best_q7 <- mean(pred_test_best_q7 != actual_test_q7)
test_error_best_tbl <- tibble::tibble(
  metric = 'Test error rate (best cutoff)',
  value = round(test_error_best_q7, 4)
)

# Print test error table.
test_error_best_tbl %>%
  table_latex(
    col_names = c('Metric', 'Value'),
    caption = 'Test error using best cutoff from training ROC.'
  )
```

Table 12: Test error using best cutoff from training ROC.

| Metric | Value |
|---|---|
| Test error rate (best cutoff) | 0.1016 |

Using the best cutoff from Problem 10, the test error is $\boxed{0.1016}$ (about 10.16%).

# Problem 12

**Problem 12**

For a student, with 0 balance, and 20000 income, what is the predicted probability of default? Use the model on training data.

Cell 14

```r
# Build student profile.
new_student_q12 <- data.frame(
  student = factor('Yes', levels = levels(default_train$student)),
  balance = 0,
  income  = 20000
)

# Compute predicted probability.
p_q12 <- as.numeric(stats::predict(model_q7, newdata = new_student_q12, type = 'response'))
q12_tbl <- tibble::tibble(
  metric = 'Predicted probability',
  value  = signif(p_q12, 6)
)

# Print probability table.
q12_tbl %>%
  table_latex(
    col_names = c('Metric', 'Value'),
    caption   = 'Predicted probability for student profile.'
  )
```

Table 13: Predicted probability for student profile.

| Metric | Value |
|---|---|
| Predicted probability | 0.0000088 |

The predicted probability is $\boxed{0.00000882}$.

# Problem 13

**Problem 13**

For a non-student, with 0 balance, and 20000 income, what is the predicted probability of default? Use the model on training data.

Cell 15

```r
# Build non-student profile.
new_nonstudent_q13 <- data.frame(
  student = factor('No', levels = levels(default_train$student)),
  balance = 0,
  income  = 20000
)

# Compute predicted probability.
p_q13 <- as.numeric(stats::predict(model_q7, newdata = new_nonstudent_q13, type = 'response'))
q13_tbl <- tibble::tibble(
  metric = 'Predicted probability',
  value  = signif(p_q13, 6)
)

# Print probability table.
q13_tbl %>%
  table_latex(
    col_names = c('Metric', 'Value'),
    caption   = 'Predicted probability for non-student profile.'
  )
```

Table 14: Predicted probability for non-student profile.

| Metric | Value |
|---|---|
| Predicted probability | 0.0000137 |

The predicted probability is $\boxed{0.00001369}$.

# Problem 14

**Problem 14**

What is the difference between the predicted values of Problems 12 and 13? Does this difference come from the coefficient of the student variable? How are they related?

Cell 16

```r
# Compute difference and student effect.
diff_q14 <- p_q12 - p_q13
beta_student_q14 <- as.numeric(stats::coef(model_q7)['studentYes'])
odds_ratio_student_q14 <- exp(beta_student_q14)

# Build relation table.
q14_tbl <- tibble::tibble(
  metric = c(
    'p(student = Yes) - p(student = No)',
    'student coefficient (log-odds scale)',
    'student odds ratio'
  ),
  value = c(diff_q14, beta_student_q14, odds_ratio_student_q14)
) %>%
  dplyr::mutate(value = signif(value, 6))

# Print relation table.
q14_tbl %>%
  table_latex(
    col_names = c('Metric', 'Value'),
    caption   = 'Probability difference and student coefficient relation.'
  )
```

Table 15: Probability difference and student coefficient relation.

| Metric | Value |
|---|---|
| p(student = Yes) - p(student = No) | -0.0000049 |
| student coefficient (log-odds scale) | -0.4398230 |
| student odds ratio | 0.6441500 |

The difference is $\boxed{p_{12} - p_{13} = -4.87 \times 10^{-6}}$, so the student profile has a slightly lower predicted default probability here. This difference is directly related to the coefficient. Let

$$r = \frac{p_{12}/(1-p_{12})}{p_{13}/(1-p_{13})} = e^{\hat{\beta}_{\texttt{studentYes}}} = 0.6442.$$

Solving this for $p_{12}$ in terms of $p_{13}$,

$$p_{12} = \frac{r\,p_{13}}{1 - p_{13} + r\,p_{13}},$$

so

$$p_{12} - p_{13} = \frac{p_{13}(1 - p_{13})(r - 1)}{1 - p_{13} + r\, p_{13}}.$$

This shows the probability difference comes from the student coefficient via $r = e^{\hat{\beta}_{\texttt{studentYes}}}$. Plugging in $r = 0.6442$ and $p_{13} = 1.368903 \times 10^{-5}$ gives

$$p_{12} - p_{13} \approx -4.87 \times 10^{-6},$$

which matches the observed difference.